

OE-EC804B — Microwave Integrated Circuits

Semester VIII • 3L:0T:0P • 3 Credits

Module 2 Exam Notes: Planar Transmission Lines-I (Stripline & Microstrip)

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1 Stripline and Microstrip Line Fundamentals

Planar transmission lines form the foundational interconnections and resonant components in MIC design.

1.1 Stripline Geometry and Pure TEM Propagation

A **Stripline** consists of a flat strip conductor of width W and thickness t centered symmetrically inside a dielectric substrate of thickness b , bounded above and below by two parallel conducting ground planes.

Stripline Pure TEM Propagation

Because the strip conductor is completely embedded inside a single, homogeneous dielectric medium (ϵ_r), the wave phase velocity is uniform throughout the cross-section. Stripline supports a **pure Transverse Electromagnetic (TEM) mode** with zero longitudinal electric ($E_z = 0$) or magnetic ($H_z = 0$) field components.

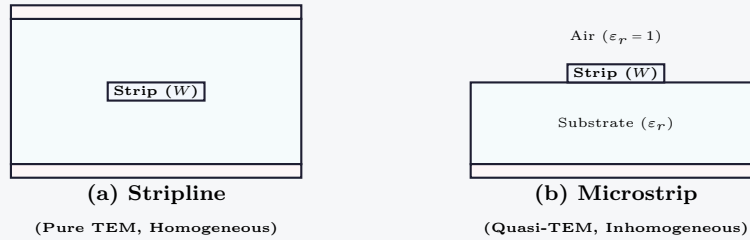
1.2 Microstrip Line Geometry and Quasi-TEM Mode

A **Microstrip Line** consists of a strip conductor of width W and thickness t printed on top of a dielectric substrate of thickness h , backed by a single ground plane on the bottom.

Quasi-TEM Mode in Microstrip

Because the top surface of a microstrip line is open to air ($\epsilon_r = 1$) while the lower region consists of dielectric ($\epsilon_r > 1$), the electromagnetic field spans two different dielectric media. The phase velocity in air (c) differs from that in the substrate ($c/\sqrt{\epsilon_r}$). Pure TEM propagation cannot be supported; the fundamental mode is an inhomogeneous **Quasi-TEM mode** with tiny longitudinal field components ($E_z, H_z \neq 0$).

Cross-Sectional Geometries and Field Distribution: Stripline vs. Microstrip



2 Effective Dielectric Constant (ϵ_{eff}) and Analysis

2.1 Concept of Effective Dielectric Constant (ϵ_{eff})

To simplify the analysis of inhomogeneous microstrip lines, the air-dielectric boundary is replaced by a fictitious homogeneous dielectric medium extending throughout space, characterized by an **Effective Dielectric Constant** (ϵ_{eff}).

Effective Dielectric Constant (ϵ_{eff}) Formula

The effective dielectric constant is bounded by $1 < \epsilon_{eff} < \epsilon_r$ and is calculated using Schneider's closed-form expression:

$$\epsilon_{eff} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left[1 + 12 \frac{h}{W} \right]^{-1/2} \quad (1)$$

where W is the microstrip width and h is the substrate thickness.

2.2 Guided Wavelength and Phase Velocity

The phase velocity v_p and guided wavelength λ_g in microstrip are given by:

$$v_p = \frac{c}{\sqrt{\epsilon_{eff}}} \quad ; \quad \lambda_g = \frac{v_p}{f} = \frac{\lambda_0}{\sqrt{\epsilon_{eff}}} \quad (2)$$

2.3 Microstrip Characteristic Impedance (Z_0) Analysis & Synthesis

1. Analysis Formulas (Given W/h and ϵ_r , find Z_0):

- For Narrow Strips ($W/h \leq 1$):

$$Z_0 = \frac{60}{\sqrt{\epsilon_{eff}}} \ln \left[8 \frac{h}{W} + 0.25 \frac{W}{h} \right] \quad (3)$$

- For Wide Strips ($W/h \geq 1$):

$$Z_0 = \frac{120\pi}{\sqrt{\epsilon_{eff}} \left[\frac{W}{h} + 1.393 + 0.667 \ln \left(\frac{W}{h} + 1.444 \right) \right]} \quad (4)$$

2. Synthesis Formulas (Given target Z_0 and ϵ_r , find required W/h):

- For $W/h < 2$:

$$\frac{W}{h} = \frac{8e^A}{e^{2A} - 2} \quad \text{where } A = \frac{Z_0}{60} \sqrt{\frac{\epsilon_r + 1}{2}} + \frac{\epsilon_r - 1}{\epsilon_r + 1} \left(0.23 + \frac{0.11}{\epsilon_r} \right) \quad (5)$$

- For $W/h > 2$:

$$\frac{W}{h} = \frac{2}{\pi} \left[B - 1 - \ln(2B - 1) + \frac{\epsilon_r - 1}{2\epsilon_r} \left(\ln(B - 1) + 0.39 - \frac{0.61}{\epsilon_r} \right) \right] \quad (6)$$

$$\text{where } B = \frac{377\pi}{2Z_0\sqrt{\epsilon_r}}.$$

2.4 Stripline Characteristic Impedance Formula

For a symmetrical stripline with ground plane spacing b and effective strip width W_e :

$$Z_{0,\text{stripline}} = \frac{30\pi}{\sqrt{\epsilon_r}} \frac{b}{W_e + 0.441b} \quad \text{where } W_e = W \text{ for } t/b \rightarrow 0 \quad (7)$$

Worked Numerical: Microstrip Design & Wavelength Calculation

Problem: A microstrip line is printed on an Alumina substrate ($\epsilon_r = 9.8$, $h = 0.635$ mm). If the width-to-height ratio is $W/h = 1.0$, calculate:

- The effective dielectric constant ϵ_{eff} .
- The characteristic impedance Z_0 .
- The guided wavelength λ_g at 10 GHz.

Solution:

- Calculate ϵ_{eff} using $W/h = 1.0$:

$$\begin{aligned} \epsilon_{eff} &= \frac{9.8 + 1}{2} + \frac{9.8 - 1}{2} [1 + 12(1)]^{-1/2} \\ &= 5.4 + 4.4 [13]^{-1/2} = 5.4 + \frac{4.4}{3.6055} \\ &= 5.4 + 1.2203 = 6.6203 \end{aligned} \quad (8)$$

(b) Since $W/h = 1.0 \leq 1$, use the narrow strip formula for Z_0 :

$$\begin{aligned} Z_0 &= \frac{60}{\sqrt{6.6203}} \ln [8(1) + 0.25(1)] \\ &= \frac{60}{2.573} \ln(8.25) = 23.319 \times 2.1102 \\ &= 49.21 \, \Omega \approx 50 \, \Omega \end{aligned} \quad (9)$$

(c) Free-space wavelength $\lambda_0 = \frac{c}{f} = \frac{3 \times 10^8 \text{ m/s}}{10 \times 10^9 \text{ Hz}} = 30 \text{ mm}$.

$$\lambda_g = \frac{\lambda_0}{\sqrt{\epsilon_{eff}}} = \frac{30 \text{ mm}}{\sqrt{6.6203}} = \frac{30}{2.573} = 11.66 \text{ mm} \quad (10)$$

3 Losses, Dispersion, and Discontinuities in Microstrip

3.1 Microstrip Attenuation & Loss Mechanisms

Total attenuation constant $\alpha_{total} = \alpha_c + \alpha_d + \alpha_r$ (Np/m or dB/m):

- **Conductor Loss (α_c):** Skin effect causes current to crowd within skin depth $\delta = \sqrt{\frac{1}{\pi f \mu \sigma}}$ along lower surface edges:

$$\alpha_c \approx \frac{8.686 R_s}{Z_0 W} \text{ dB/m} \quad \text{where } R_s = \sqrt{\frac{\pi f \mu}{\sigma}} \text{ (Surface Resistance)} \quad (11)$$

- **Dielectric Loss (α_d):** Caused by substrate dipole relaxation loss:

$$\alpha_d = 27.3 \frac{\epsilon_r}{\sqrt{\epsilon_{eff}}} \frac{\epsilon_{eff} - 1}{\epsilon_r - 1} \frac{\tan \delta}{\lambda_0} \text{ dB/m} \quad (12)$$

- **Radiation Loss (α_r):** At high frequencies, microstrip open boundaries act as long slot radiators emitting RF energy into space.

3.2 Microstrip Discontinuities

Physical transitions in microstrip layouts alter local electric and magnetic field distributions, generating parasitic reactances.

Discontinuity Type	Physical Origin	Equivalent Circuit Model & Mitigation
Open-End Line	Fringing electric fields extend past strip termination into air	Shunt end capacitance C_{oc} , equivalent to extra length $\Delta l = 0.412h \frac{\epsilon_{eff} + 0.3}{\epsilon_{eff} - 0.258} \frac{W/h + 0.264}{W/h + 0.8}$
Step-Change in Width	Sudden width transition $W_1 \rightarrow W_2$ disrupts current distribution	Series inductance L_s and shunt capacitance C_s .
Right-Angle Bend (90°)	Current crowding at inner corner; excess charge at outer corner	Series inductance L_b and shunt capacitance C_b . Mitigated by 45° Mitred Bend ($M = 52 + 65e^{-1.35W/h}\%$ metal removal).
T-Junction / Cross	Junction of orthogonal microstrip branches	Equivalent T-network of series inductances and shunt capacitance.

Quick Revision — Module 2

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Key Concepts and Metrics at a Glance

Effective Dielectric ϵ_{eff}	$\epsilon_{eff} = \frac{\epsilon_r+1}{2} + \frac{\epsilon_r-1}{2} \left[1 + 12 \frac{h}{W}\right]^{-1/2} \quad (1 < \epsilon_{eff} < \epsilon_r)$
Phase Velocity & Wavelength	$v_p = \frac{c}{\sqrt{\epsilon_{eff}}} \quad ; \quad \lambda_g = \frac{\lambda_0}{\sqrt{\epsilon_{eff}}}$
Stripline Propagation	Pure TEM mode ($E_z = 0, H_z = 0$) due to symmetrical homogeneous dielectric.
Microstrip Propagation	Inhomogeneous Quasi-TEM mode ($E_z, H_z \neq 0$) due to air-substrate boundary.
Mitred Bend Optimal Cut	$M = \frac{x}{d\sqrt{2}} \times 100\% = 52 + 65e^{-1.35W/h}\%$ removes 90° corner capacitance.

One-Line Definitions

- **Quasi-TEM Mode:** Hybrid electromagnetic wave in non-homogeneous media possessing tiny longitudinal field components.
- **Fringing Fields:** Electric field lines extending into the air region surrounding microstrip edges.
- **Mitred Bend:** A 45° chamfered microstrip bend removing excess corner capacitance to maintain 50 Ω matching.
- **Microstrip Dispersion:** Frequency-dependent variation of $\epsilon_{eff}(f)$ causing phase velocity to decrease at high frequencies.

Frequently Asked / Viva Questions

Q1. Why does a microstrip line support a Quasi-TEM mode instead of a pure TEM mode?

Because the electromagnetic field exists in two different media (dielectric substrate and air), causing phase velocity mismatch that induces small longitudinal E_z and H_z components.

Q2. Why does a stripline support a pure TEM mode?

Because the conductor is completely enclosed inside a single homogeneous dielectric bounded by two ground planes.

Q3. What is the Effective Dielectric Constant (ϵ_{eff}) of a microstrip?

A weighted average dielectric constant of air and substrate representing a fictitious homogeneous medium supporting the same wave velocity.

Q4. How does characteristic impedance (Z_0) vary with strip width W ?

As strip width W increases, capacitance increases and characteristic impedance Z_0 decreases ($Z_0 \propto 1/W$).

Q5. Explain the physical significance of an open-end microstrip discontinuity.

Fringing fields at the open termination store electric energy, behaving as a shunt capacitor C_{oc} that increases the line's effective length by Δl .

Q6. Why are right-angle microstrip bends mitred (chamfered)?

A 90° sharp bend creates excess capacitance at the outer corner. Mitring removes 50% to 60% of the corner metal, eliminating reflections.

Q7. What are the main loss mechanisms in microstrip lines?

Frequently Asked / Viva Questions (contd.)

Conductor loss α_c (skin effect), dielectric loss α_d (substrate dissipation), and radiation loss α_r (open boundary power leakage).

Q8. Differentiate between Stripline and Microstrip in terms of isolation.

Stripline provides superior shielding against crosstalk due to dual ground planes. Microstrip suffers higher radiation loss due to its open top boundary.

Q9. How does dispersion affect microstrip performance at high frequencies?

As frequency increases, fields become more confined inside the substrate, increasing $\varepsilon_{eff}(f)$ towards ε_r and causing dispersion (signal distortion).

Q10. What is the typical value of ε_{eff} relative to ε_r ?

ε_{eff} is strictly bounded by $1 < \varepsilon_{eff} < \varepsilon_r$, typically around $0.65\varepsilon_r$ for standard microstrips.